Lecture 8: Planetary Interiors
Structure, Composition, & Dynamics

Planetary Systems

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Earth's interior cross-section. Credit: NASA/JPL-Caltech

Learning objectives

By the end of this lecture, you will be able to:

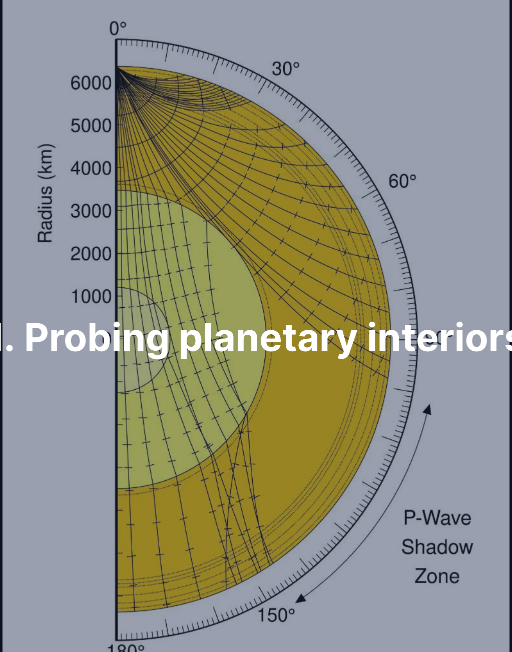
- Explain how seismology, gravity, and the moment of inertia probe planetary interiors
- Derive C/MR^2 for uniform and two-layer bodies
- Apply hydrostatic equilibrium and equations of state at planetary conditions
- Compare the internal structures of terrestrial planets, giant planets, and icy moons
- Describe recent results from InSight, Juno, and Europa Clipper

Recap: Lecture 7

- Impact cratering: morphology, scaling laws, crater counting
- Volcanism: effusive vs. explosive, shield volcanoes, flood basalts
- Tectonics: extensional, compressional, strike-slip faulting
- Erosion and weathering: wind, water, ice, space weathering
- Cryovolcanism on icy bodies (Enceladus, Triton)

Today: What lies *beneath* the surface? How do we know, and what does it tell us?

1. Probing planetary interiors



Seismic P-wave paths through Earth. Credit: Wikimedia Commons, public domain

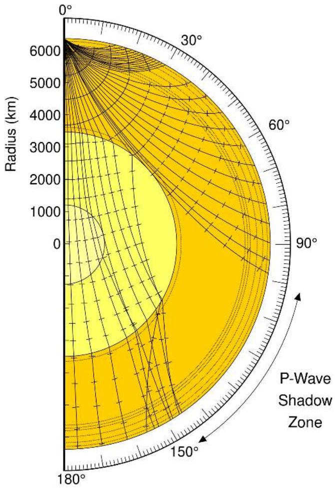
Three ways to see inside a planet

We cannot directly sample deep interiors. Three complementary tools:

1. **Seismology:** wave propagation through layers of different density and rigidity
2. **Gravity field measurements:** mass distribution from spacecraft tracking
3. **Moment of inertia:** degree of central mass concentration

For Earth: all three are available at high precision.
For other planets: gravity and C/MR^2 are often all we have.

Seismic waves: P-waves and S-waves



Credit: Wikimedia Commons, public domain

P-waves (primary, compressional):

- Travel through solids *and* liquids
- $v_P = \sqrt{(K + \frac{4}{3}\mu)/\rho}$

S-waves (secondary, shear):

- Travel through solids *only*
- $v_S = \sqrt{\mu/\rho}$
- $\mu = 0$ in a liquid $\Rightarrow$ S-waves **cannot propagate**

The shadow zone: proof of a liquid core

- At the core-mantle boundary v_P drops from 13.7 to 8.1 km s^{-1}, so P-waves entering the core are bent sharply downward
- Result: a **P-wave shadow zone** with no direct arrivals from ~98° to ~145° (classically quoted as 103° to 143°)
- S-waves cannot cross a liquid at all: **no direct S beyond** ~103°
- Discovered by Oldham (1906) and Gutenberg (1914)

The absence of S-waves through the outer core is the definitive evidence that it is liquid.

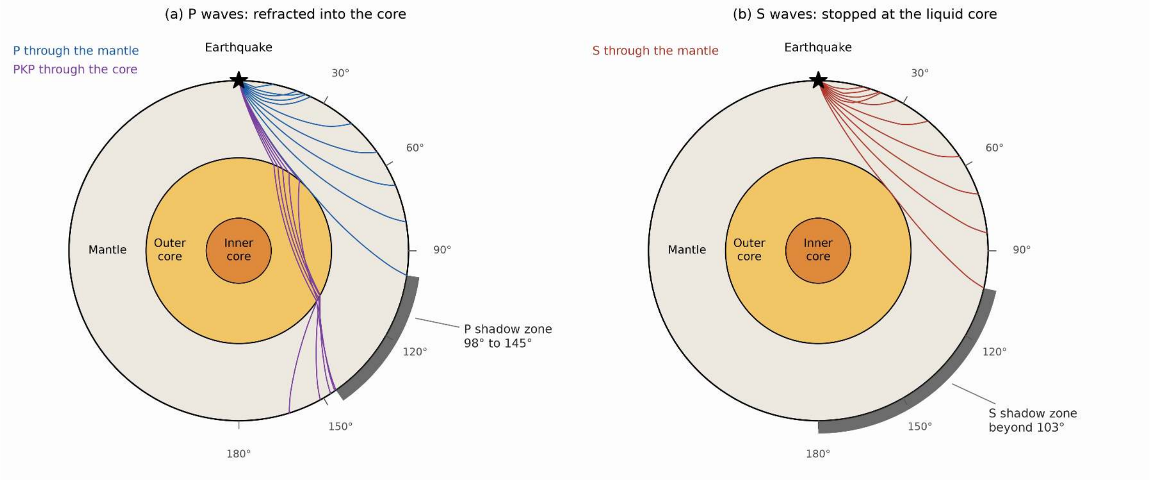
Why the rays bend: one conserved quantity

In a spherically symmetric body the **ray parameter** is constant along a ray:

$$p = \frac{r \sin i}{v(r)}$$

where i is the angle to the local vertical.

- A ray bottoms out where $r/v(r) = p$
- Angular distance: $\Delta = 2\int_{r_{\mathrm{turn}}}^{R} \frac{p\,\mathrm{d}r}{r\sqrt{(r/v)^2 - p^2}}$
- Feed PREM into this integral and both shadow zones fall out, with no free parameters



Rays traced through PREM. Faint PKIKP arrivals through the solid inner core return into the P shadow from ~116°. Course original, from the PREM tabulation of Dziewonski & Anderson (1981).

Gravity field measurements

- A perfectly spherical, homogeneous body produces a pure $1/r^2$ field
- Real planets are oblate due to rotation $\Rightarrow$ **gravitational quadrupole** J_2 (deviation from a pure $1/r^2$ field)
- J_2 depends on mass distribution between equator and poles
- Higher harmonics reveal lateral density variations
- Measured via spacecraft radio tracking (Doppler shifts)
- Examples: GRAIL (Moon), Juno (Jupiter), MRO (Mars)

Precision: Juno measures Jupiter's gravity harmonics to parts in 10^9.

The moment of inertia factor: C/MR^2

The **most diagnostic single number** for internal mass distribution, i.e. how strongly a body has separated into a dense core and light mantle:

Body	C/MR^2	Interpretation
Uniform sphere	0.400	Homogeneous, undifferentiated
Moon	0.393	Weakly differentiated, small core
Mars	0.364	Moderately differentiated
Mercury	0.346	Large dense core
Earth	0.331	Strongly differentiated, iron core

Source: de Pater & Lissauer (2010), Table 6.1

Determined from J_2 + spin-axis precession rate (or **libration** amplitude, a small periodic rocking about the mean orientation, for tidally locked bodies), combined via the **Darwin–Radau relation** (links C/MR^2 to J_2 and flattening).

Measuring C/MR^2 in practice

Two observables needed:

1. **Gravitational oblateness** J_2 from spacecraft tracking
2. **Precession rate** of the spin axis (or forced libration for synchronous rotators)

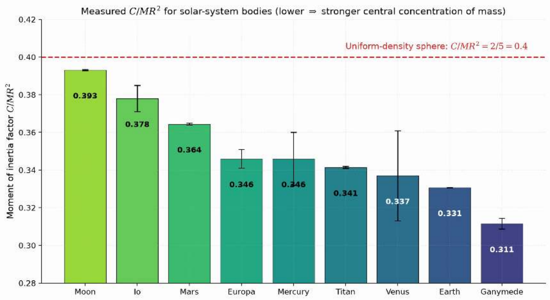
These combine via the **Darwin–Radau relation**:

$$\frac{C}{MR^2} = \frac{2}{3}\left[1 - \frac{2}{5}\sqrt{\frac{4 - k_f}{1 + k_f}}\right]$$

where k_f is the fluid Love number (a measure of the body's response to rotation).

Result: a single number that immediately distinguishes a rubble pile (0.4) from a differentiated body like Earth (0.33).

One number per body

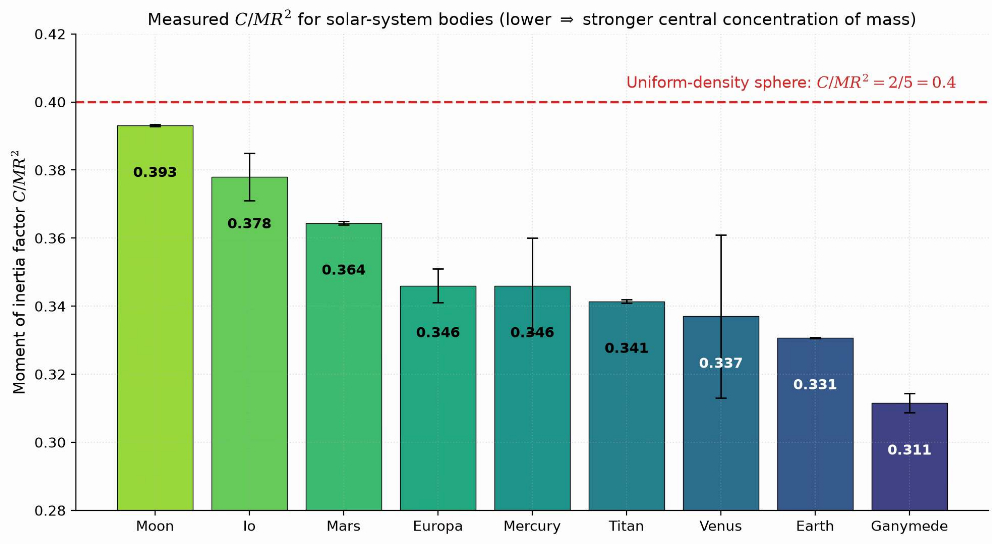


Credit: Course original; lunar value from Williams et al. (2014)

Measured C/MR^2 across the solar system:

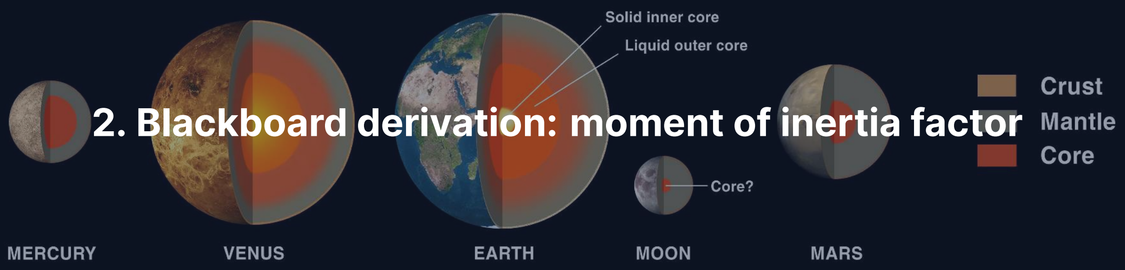
- Every body falls below the uniform-sphere line at $2/5 = 0.400$: all are centrally condensed
- The Moon (0.393) is only weakly differentiated; Earth (0.331) and Ganymede (0.311) strongly so
- Values follow from J_2 plus precession or libration via the Darwin–Radau relation

C/MR^2 is often the first interior constraint available for a newly characterised body.



Moment of inertia factors for nine solar-system bodies. Course original; lunar value from Williams et al. (2014).

2. Blackboard derivation: moment of inertia factor



Differentiated interiors of the terrestrial bodies. Credit: NASA Solar System Exploration, public domain

Deriving C/MR^2

Goal: Derive the moment of inertia factor for (a) a uniform sphere and (b) a two-layer differentiated body. Show how the measured value constrains core size.

Starting point: For a spherically symmetric body, the moment of inertia about the rotation axis is:

$$C = \frac{8\pi}{3} \int_0^R \rho(r)\, r^4 \, \mathrm{d}r$$

This comes from summing $\frac{2}{3} r^2 \, \mathrm{d}m$ over thin spherical shells.

Step 1: uniform sphere

Set ρ = const:

$$C = \frac{8\pi}{3}\rho \int_0^R r^4 \, \mathrm{d}r = \frac{8\pi}{3}\rho \frac{R^5}{5} = \frac{8\pi}{15}\rho R^5$$

Total mass: $M = \frac{4}{3}\pi R^3 \rho$

$$\frac{C}{MR^2} = \frac{8\pi\rho R^5/15}{(4\pi\rho R^3/3)\,R^2} = \frac{2}{5} = 0.400$$

This is the **maximum** for any body with $\rho(r)$ that does not increase outward.

Step 2: two-layer body

Dense core (ρ_c, radius R_c) + lighter mantle (ρ_m, radius R):

$$C = \frac{8\pi}{15}\left[(\rho_c - \rho_m)\,R_c^5 + \rho_m\,R^5\right]$$

$$M = \frac{4}{3}\pi\left[(\rho_c - \rho_m)\,R_c^3 + \rho_m\,R^3\right]$$

Define $x \equiv R_c/R$ and $f \equiv \rho_c/\rho_m$:

$$\frac{C}{MR^2} = \frac{2}{5}\cdot\frac{(f-1)\,x^5 + 1}{(f-1)\,x^3 + 1}$$

Since $x^5 < x^3$ for $0 < x < 1$, the ratio is < 1 when $f > 1$.
More mass in the centre ⇒ lower C/MR^2.

Worked example: Earth

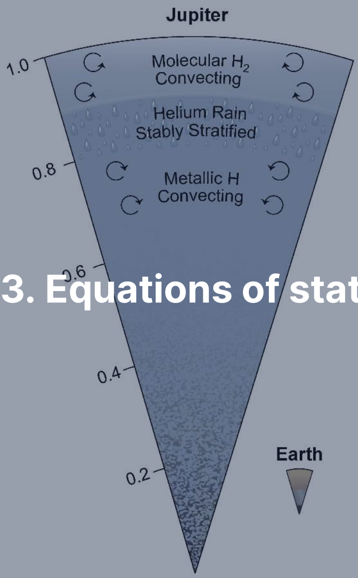
Parameter	Symbol	Value
Core radius fraction	$x = R_c/R$	$3480/6371 = 0.546$
Density ratio	$f = \rho_c/\rho_m$	$11{,}000/4{,}400 \approx 2.5$

$$\frac{C}{MR^2} = \frac{2}{5} \times \frac{1.5 \times 0.049 + 1}{1.5 \times 0.163 + 1} = 0.4 \times \frac{1.073}{1.244} \approx 0.345$$

Measured value: $C/MR^2 = 0.3307$

Remaining difference: continuous density increase within each layer, especially the ~30% jump from outer to inner core.

3. Equations of state



Jupiter's interior. Credit: NASA/JPL-Caltech/SwRI/John E. Connerney (PIA25062)

Hydrostatic equilibrium in spherical symmetry

Pressure at each depth supports the weight of overlying material:

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\rho(r)\,g(r) = -\frac{G\,m(r)\,\rho(r)}{r^2}$$

where $m(r) = 4\pi \int_0^r \rho(r')\,r'^2\,\mathrm{d}r'$ is the mass enclosed within radius r.

Combined with an equation of state $P = P(\rho, T)$, this can be integrated inward from the surface to determine the full pressure–density profile.

Central pressure estimate

For uniform density $\bar{\rho} = 3M/(4\pi R^3)$:

$$P_c \approx \frac{3\,G\,M^2}{8\pi\,R^4}$$

Body	**Estimated P_c**	**Actual P_c**
Earth	~170 GPa	~360 GPa
Mars	~25 GPa	~40 GPa
Jupiter	~700 GPa	~4000 GPa

The uniform-density estimate is always **too low** because real bodies have higher density at the centre.

Birch's law

Birch (1952): empirical linear relationship between compressional wave velocity v_P and density ρ for silicates and oxides:

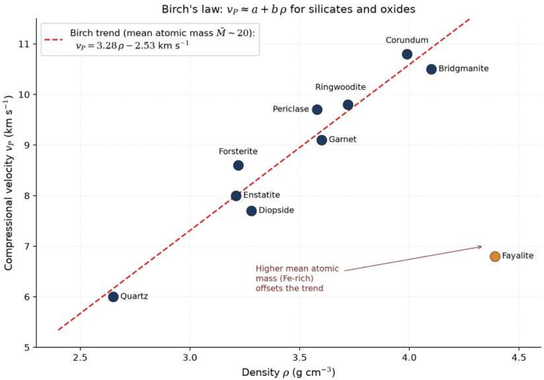
$$v_P = a(\bar{M}) + b\,\rho$$

where $\bar{M}$ is the mean atomic weight and a, b are empirical constants.

- Connects seismology directly to density
- Holds remarkably well for mantle minerals at high pressure
- Breaks down at phase transitions and compositional boundaries

Birch's law is a cornerstone of mineral physics, linking lab measurements, seismic data, and interior models.

Birch's law in the laboratory

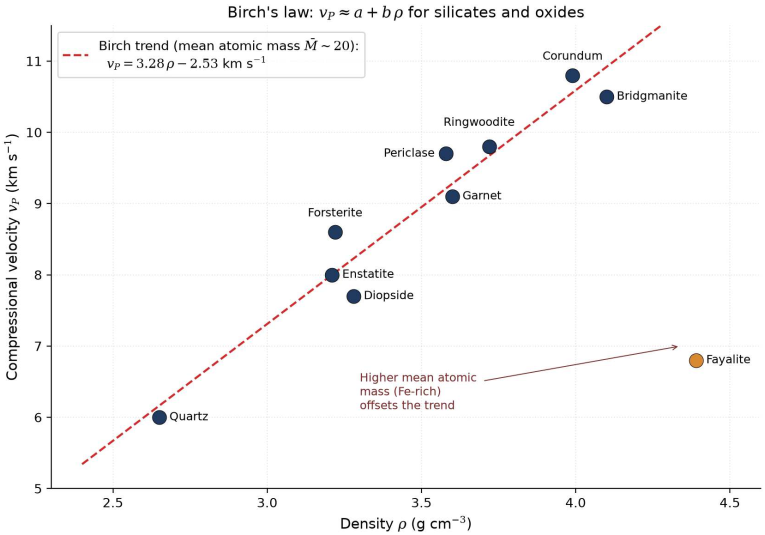


Credit: Course original; trend after Birch (1952)

v_P against density for mantle-forming minerals:

- The trend is approximately linear at fixed mean atomic mass
- It maps seismic velocities measured at depth to densities, provided the composition is known
- Fe-rich outliers such as fayalite sit below the trend: higher mean atomic mass lowers v_P at a given density

Birch's law turns velocity profiles into density profiles: the bridge from seismology to composition.



Birch's law: v_P against density for mantle minerals. Course original; trend after Birch (1952).

Adams–Williamson equation

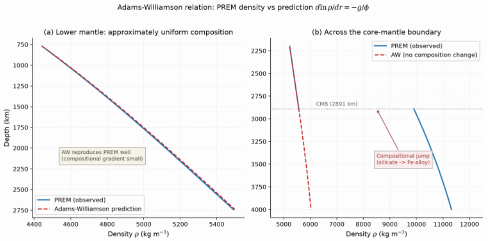
The **seismic parameter**: $\phi \equiv K_S/\rho = v_P^2 - \frac{4}{3}v_S^2$ (K_S: bulk modulus, resistance to compression)

Combining with hydrostatic equilibrium:

$$\frac{\mathrm{d}\rho}{\mathrm{d}r} = -\frac{\rho(r)\,g(r)}{\phi(r)}$$

- Determines density profile from measured seismic wave speeds
- Valid where composition is approximately uniform
- Breaks down at phase transitions and compositional boundaries
- Foundation of the PREM density model

Testing Adams–Williamson against PREM

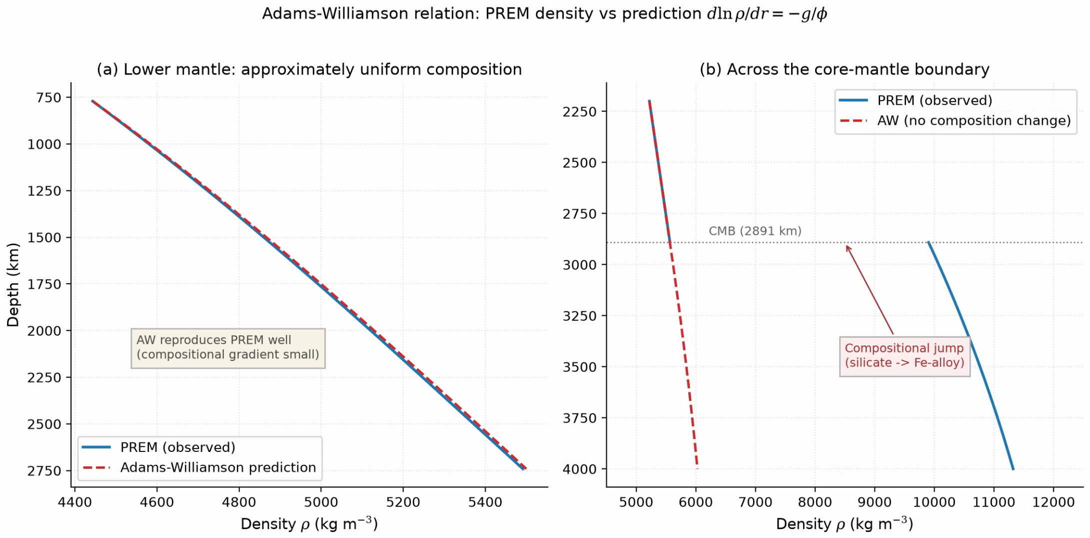


Credit: Course original, from the PREM tabulation

The equation against the real Earth:

- In the lower mantle (771–2741 km) the integrated profile reproduces PREM: the density rise is self-compression alone
- At the CMB the observed density jumps from ~5570 to ~9900 kg m^{-3}; the extrapolation predicts a smooth, far smaller increase
- The deviation exposes the compositional change from silicate mantle to iron-alloy core

Where Adams–Williamson breaks down is as informative as where it works: discontinuities mark composition changes.

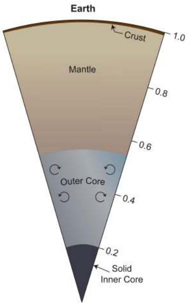


The Adams–Williamson relation tested against PREM. Course original, from the PREM tabulation of Dziewonski & Anderson (1981).

4. Earth’s interior

Earth’s interior. Credit: NASA/JPL-Caltech/SwRI/John E. Connerney (PIA25063)

Earth's internal structure



Credit: NASA/JPL-Caltech

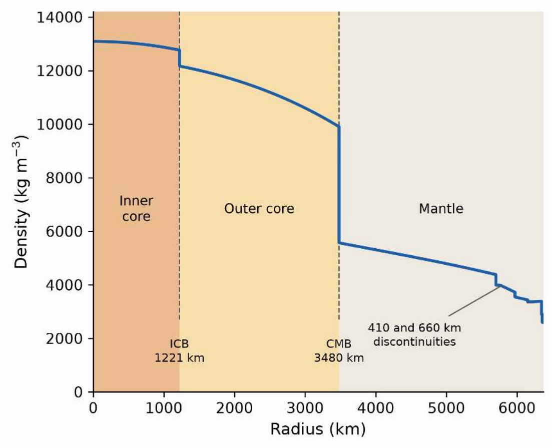
Four major layers:

1. **Crust** (0–5–70 km)
2. **Mantle** (Moho–2891 km)
3. **Outer core** (2891–5150 km): liquid
4. **Inner core** (5150–6371 km): solid

The inner core is solid **because of pressure**: at 330–360 GPa, iron's melting point exceeds the core's 5000–6000 K.

Mean density	5515 kg m^{-3}
C/MR^2	0.3307
Surface temp	~288 K
Central temp	~5000–6000 K
Central pressure	~360 GPa

The PREM density profile



Credit: Course original; PREM coefficients from Dziewonski & Anderson (1981)

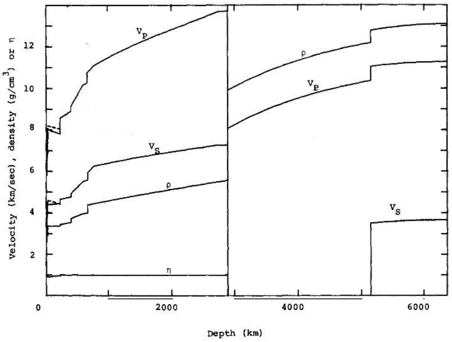
Preliminary Reference Earth Model (PREM):

- Upper mantle: ~3300 $kg\,m^{-3}$
- Lower mantle base: ~5500 $kg\,m^{-3}$
- Outer core: ~9900–12,200 $kg\,m^{-3}$
- Inner core: ~13,000 $kg\,m^{-3}$

Discontinuities at 410 and 660 km: mineral phase transitions.

Sharp jump at the core–mantle boundary (CMB, ~3480 km): silicate → iron alloy.

PREM: the full radial picture

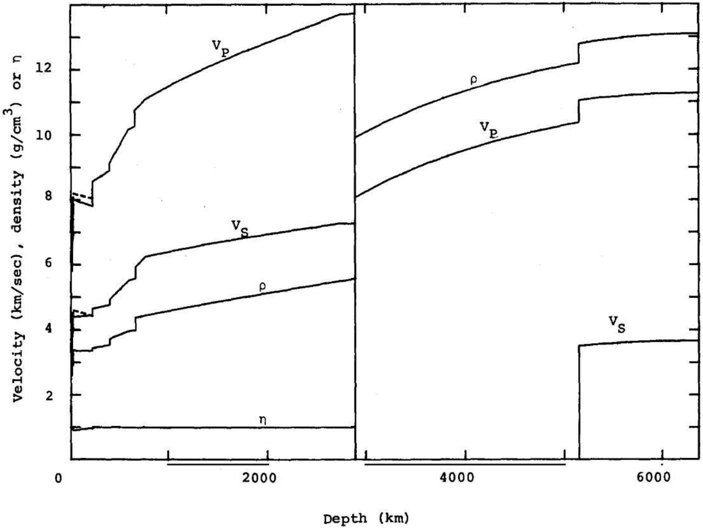


Credit: Dziewonski & Anderson (1981), Fig. 8

One model, four profiles:

- v_P, v_S, density, and anisotropy η versus depth, synthesised from millions of travel-time observations
- v_S vanishes in the outer core and resumes in the inner core: a liquid shell around a solid centre
- Fine structure in the uppermost mantle and transition zone: the 410 and 660 km discontinuities

PREM is the reference model against which every other planetary interior is measured.



PREM radial profiles of wave speeds, density, and anisotropy. Credit: Dziewonski & Anderson (1981), Fig. 8.

The crust

Oceanic crust:

- 5–10 km thick
- Basaltic, $\rho \approx 3000\ \mathrm{kg\,m^{-3}}$
- Young: <200 Myr (recycled by subduction)

Continental crust:

- 30–70 km thick
- Granitic, $\rho \approx 2700\ \mathrm{kg\,m^{-3}}$
- Old: up to ~4 Gyr

The crust is <1% of Earth's mass, but contains a large fraction of incompatible and heat-producing elements (U, Th, K).

Base of the crust = the **Mohorovičić discontinuity** (Moho), discovered 1909.

Mantle and core: key numbers

Layer	Depth (km)	ρ (kg m^{-3})	Composition
Upper mantle	0–410	3300–3700	Olivine, pyroxene
Transition zone	410–660	3700–4400	Wadsleyite, ringwoodite
Lower mantle	660–2891	4400–5500	Bridgmanite, ferropericlase
Outer core	2891–5150	9900–12,200	Liquid Fe–Ni + light elements
Inner core	5150–6371	~13,000	Solid Fe–Ni

Source: PREM (Dziewonski & Anderson 1981); Stixrude & Lithgow-Bertelloni (2005)

- Outer core is liquid: no S-waves, drives the **geodynamo** (fluid motion generating the magnetic field)
- Inner core grows at ~0.5 mm yr^{-1} as Earth cools
- Latent heat from inner core crystallisation helps power the dynamo (Lecture 4)

5. Mantle rheology & convection

Global surface heat flow, the surface expression of mantle convection. Credit: Fabio Crameri, Wikimedia Commons, CC BY-SA 4.0

Solid-state creep

The mantle is **solid** (transmits S-waves), yet it flows over geological timescales because its **viscosity** (resistance to flow) is finite.

- At $T > 0.5\,T_{\mathrm{melt}}$: atoms migrate along grain boundaries and through the lattice
- **Diffusion creep:** viscosity depends on grain size, dominant in lower mantle
- **Dislocation creep:** viscosity depends on stress, dominant in upper mantle

Material	Viscosity (Pa s)
Water	10^{-3}
Glacial ice	10^{12}
Upper mantle	10^{20}–10^{21}
Lower mantle	10^{22}–10^{23}

Source: Karato (2008), *Deformation of Earth Materials*

The Rayleigh number

Governed by the **Rayleigh number** (buoyancy vs. viscosity):

$$\mathrm{Ra} = \frac{\alpha\, \rho\, g\, \Delta T\, d^3}{\kappa\, \eta}$$

Parameter	Symbol	Earth value
Thermal expansivity	α	3×10^{-5} K^{-1}
Density	ρ	4000 kg m^{-3}
Gravity	g	10 m s^{-2}
Temperature contrast	ΔT	2500 K
Mantle thickness	d	2.9×10^{6} m
Thermal diffusivity	κ	10^{-6} m^2 s^{-1}
Viscosity	η	10^{21} Pa s

$\mathrm{Ra} \sim 10^7$–$10^8 \gg \mathrm{Ra}_c \approx 10^3 \Rightarrow$ **vigorous convection**.

Convection patterns: an open debate

Whole-mantle convection:

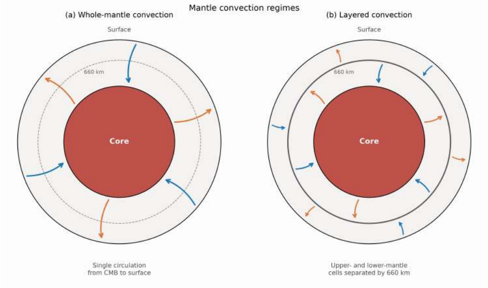
- Material circulates from surface to CMB
- Slabs penetrate through 660 km
- **Plumes** (hot upwellings) rise from CMB to surface
- Supported by seismic tomography (3D imaging) of some subducting slabs

Layered convection:

- 660 km discontinuity impedes flow
- Separate upper and lower cells
- Some slabs stagnate at 660 km
- Different chemical reservoirs persist

Reality is likely a **mixed regime**: some slabs penetrate, others stagnate. The 660 km boundary is a partial barrier, not an absolute one.

Whole-mantle or layered?

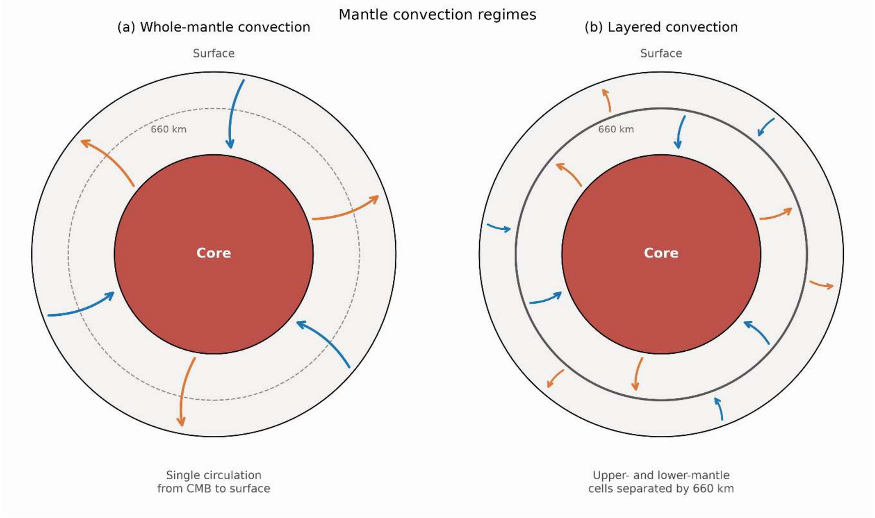


Credit: Schematic; not to scale

The two end-member regimes:

- (a) Whole-mantle: a single circulation carries material from the CMB to the surface; the 660 km boundary only weakly impedes flow
- (b) Layered: separate cells above and below 660 km, decoupled by the negative Clapeyron slope
- Seismic tomography of subducting slabs shows both behaviours

The 660 km phase boundary sets the style of mantle convection.

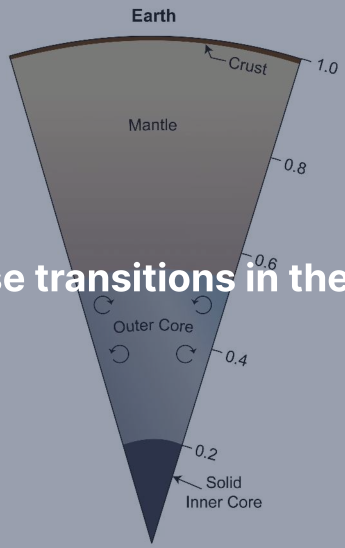


End-member regimes of mantle convection. The 660 km boundary is dashed where flow crosses it, solid where it blocks flow. Schematic; boundary radius exaggerated.

Break

Earthrise, Apollo 8 (1968). Credit: NASA

6. Phase transitions in the mantle



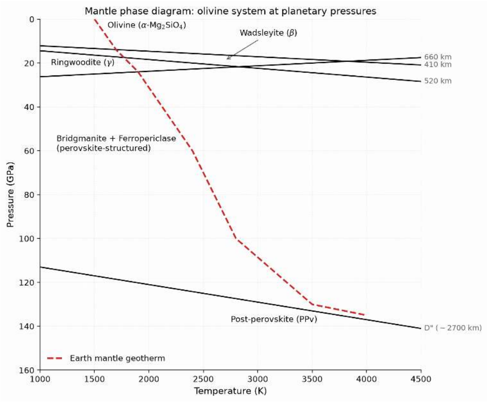
Earth's interior. Credit: NASA/JPL-Caltech/SwRI/John E. Connerney (PIA25063)

Mineral phase transitions with depth

Depth (km)	Transition	$\Delta\rho$
410	Olivine → wadsleyite	~5%
520	Wadsleyite → ringwoodite	~2%
660	Ringwoodite → bridgmanite + ferropericlase	~8%
~2700	Bridgmanite → post-perovskite	~1–2%

- Same chemical composition ($(\mathrm{Mg}, \mathrm{Fe})_2\mathrm{SiO}_4$), different crystal structures
- Driven by pressure: atoms rearrange into denser packing
- These transitions produce the **seismic discontinuities** in the PREM model

The mantle's phase map

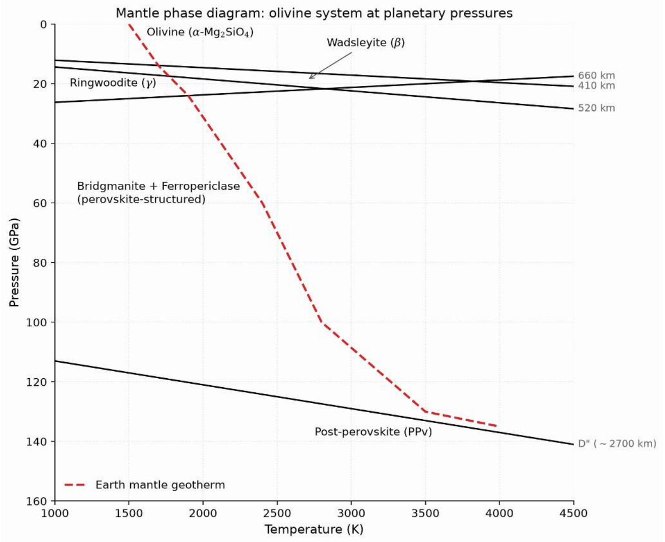


Credit: After Turcotte & Schubert (2002); Murakami et al. (2004)

Pressure–temperature map of the mantle:

- Olivine → wadsleyite at ~14 GPa (410 km), → ringwoodite at ~18 GPa (520 km), → bridgmanite + ferropericlase at ~24 GPa (660 km)
- Bridgmanite → post-perovskite at ~125 GPa: the D" layer near the CMB
- The geotherm crosses each boundary at the depth of the corresponding seismic discontinuity

Seismic discontinuities are where the geotherm crosses mineral phase boundaries.



Phase boundaries of the mantle against an approximate geotherm. After Turcotte & Schubert (2002) and Murakami et al. (2004).

The 660 km discontinuity

The most important internal boundary within the mantle:

- Ringwoodite breaks down into bridgmanite + ferropericlase
- Largest density jump (~8%) within the mantle
- **Negative Clapeyron slope:** boundary pressure *decreases* with increasing temperature
- A descending hot slab shifts the boundary *deeper*, creating buoyancy resistance
- Partially impedes convective flow: key to the layered vs. whole-mantle debate

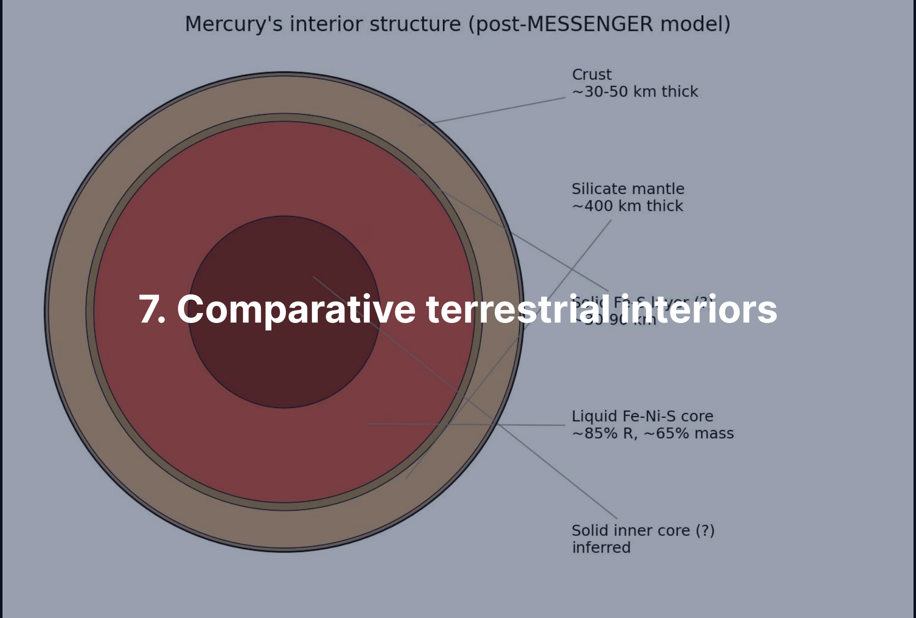
The sign of the Clapeyron slope determines whether a phase transition promotes or resists convective penetration.

Post-perovskite and the D” layer

- Discovered in lab experiments in 2004 (Murakami et al.)
- Bridgmanite → post-perovskite (a denser crystal structure) at ~125 GPa, ~2500 K
- Occurs ~200 km above the CMB (the **D” layer**)
- May explain:
  - Complex seismic velocity anomalies in D”
  - Lateral variations in the lowermost mantle
  - Anisotropy observed near the CMB

Not unique to Earth: any sufficiently large rocky body will experience these transitions at appropriate pressures.
Mars likely has an olivine–wadsleyite transition; the Moon is too small.

7. Comparative terrestrial interiors



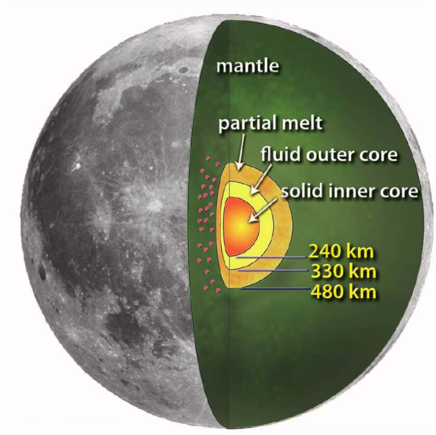
Mercury’s layered interior. After Margot et al. (2018)

The Moon

- C/MR^2 = 0.393: nearly undifferentiated (close to 0.4)
- $\bar{\rho}$ = 3346 kg m^{-3}
- Crust: ~34–43 km (anorthosite from magma ocean flotation)
- Silicate mantle to ~1400 km depth
- Small, partially molten core: R_c ~ 200–380 km (~10–20% of lunar radius)
  Lower bound from GRAIL + lunar laser ranging, upper bound from Apollo seismic reanalysis

- Apollo seismometers (1969–1977): first extraterrestrial seismology
- Near-uniform structure + low mean density consistent with formation from silicate mantle material (giant impact origin, Lecture 2)

The Moon's small core

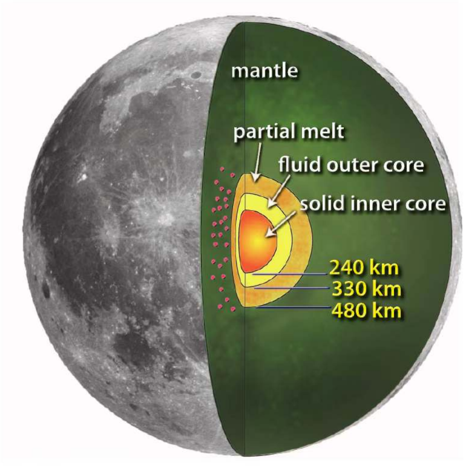


Credit: Weber et al. (2011), Fig. 1A

Apollo seismograms, reanalysed:

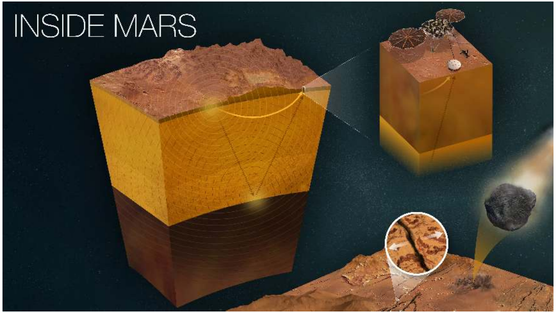
- A partial-melt boundary layer at ~480 km radius, a liquid outer core at ~330 km, and a solid inner core at ~240 km
- Red dots: the deep-moonquake source regions used to probe the deepest interior
- The moment of inertia from GRAIL and lunar laser ranging is consistent with this layering

Apollo-era data, reanalysed with modern methods, still anchor the picture of the lunar interior.



The lunar interior from Apollo seismology. Credit: Weber et al. (2011), Fig. 1A.

Mars: revealed by InSight

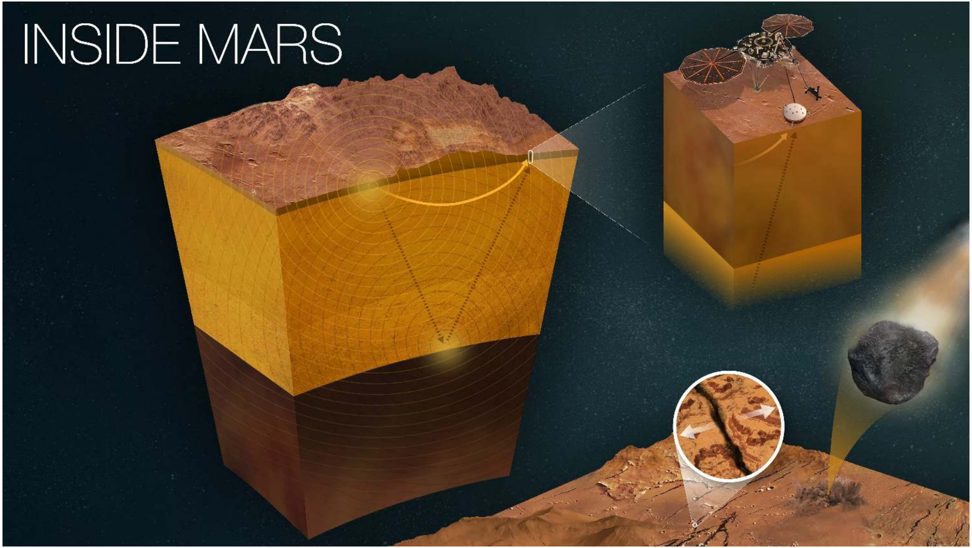


Credit: NASA/JPL-Caltech (PIA25282)

NASA InSight (2018–2022), SEIS seismometer:

- **Crust:** 24–72 km thick
- **Mantle:** to ~1560 km depth
- **Liquid core:** R_c ~ 1830 km (~54% of R; Stähler et al. 2021)
- Core density ~6000 kg m^{-3} (Earth: ~10,900): rich in sulphur and light elements
- **2023 revision:** a molten silicate layer tops the core; iron core R_c ~ 1650–1675 km, ρ ~ 6470–6650 kg m^{-3}

No active dynamo today: core not convecting vigorously enough (Lecture 4).



Mars's interior as revealed by InSight seismology. Credit: NASA/JPL-Caltech (PIA25282).

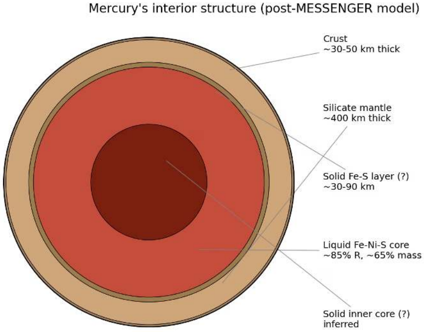
Mercury: an iron world

- $C/MR^2 = 0.346$, $\bar{\rho} = 5427\ \mathrm{kg\,m^{-3}}$
- Highest **uncompressed density** (density corrected for self-compression) of any terrestrial planet
- **Enormous iron core:** R_c ~ 2000–2050 km, ~83% of the planet's radius
- Core contains ~74% of Mercury's total mass
- Thin silicate mantle (~400 km) + crust (~30–50 km)
- Weak but global magnetic field ⇒ partly liquid, convecting core

MESSENGER spacecraft (2011–2015) + Earth-based radar libration measurements provided these constraints.

Open question: Why such a large core fraction? (Lecture 10)

Inside the iron world

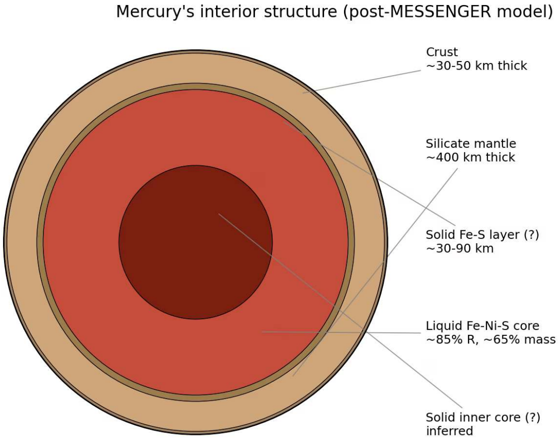


Credit: After Margot et al. (2018)

Mercury's layering:

- The metallic core reaches ~83% of the radius; a ~400 km mantle and ~30–50 km crust sit on top
- A thin solid FeS layer may cap the liquid Fe-Ni-S outer core that drives the weak magnetic field
- Radar libration measurements (Margot et al. 2007) proved a molten layer; MESSENGER fixed the dimensions

Mercury's core holds ~74% of its mass, the largest core fraction of any terrestrial planet.



Mercury’s layered interior from MESSENGER gravity and libration data. After Margot et al. (2018).

Comparing terrestrial body interiors

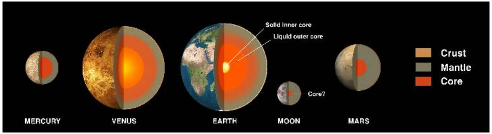
Body	$\bar{\rho}$ (kg m^{-3})	C/MR^2	R_c/R	Dynamo?
Moon	3346	0.393	~0.1–0.2	No (ancient)
Mars	3934	0.364	~0.49–0.54	No (ancient)
Venus	5243	~0.33	~0.55	No
Earth	5515	0.331	0.55	Yes
Mercury	5427	0.346	~0.83	Weak

Mars: upper value from Stähler et al. (2021), lower from the 2023 core-radius revision; Mercury: Margot et al. (2018)

Trends:

- Higher mean density ⇒ larger iron core fraction (generally)
- Lower C/MR^2 ⇒ stronger differentiation
- Active dynamo requires liquid conducting region + vigorous convection

Rocky interiors side by side

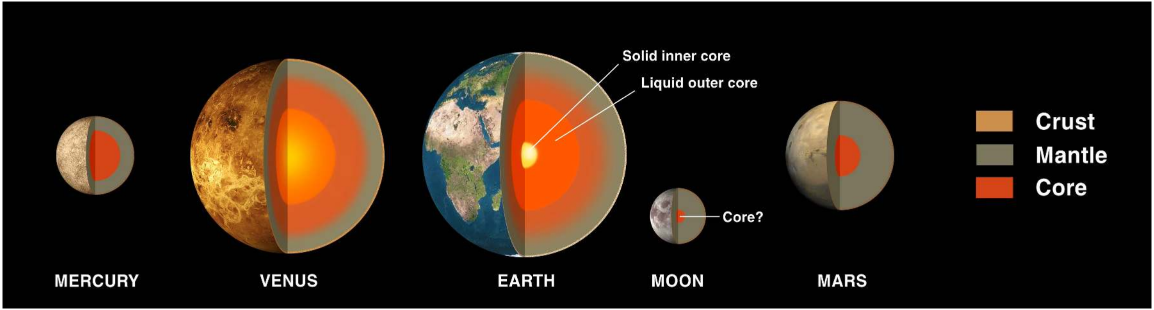


Credit: NASA Solar System Exploration, public domain

The four terrestrial planets and the Moon at relative scale:

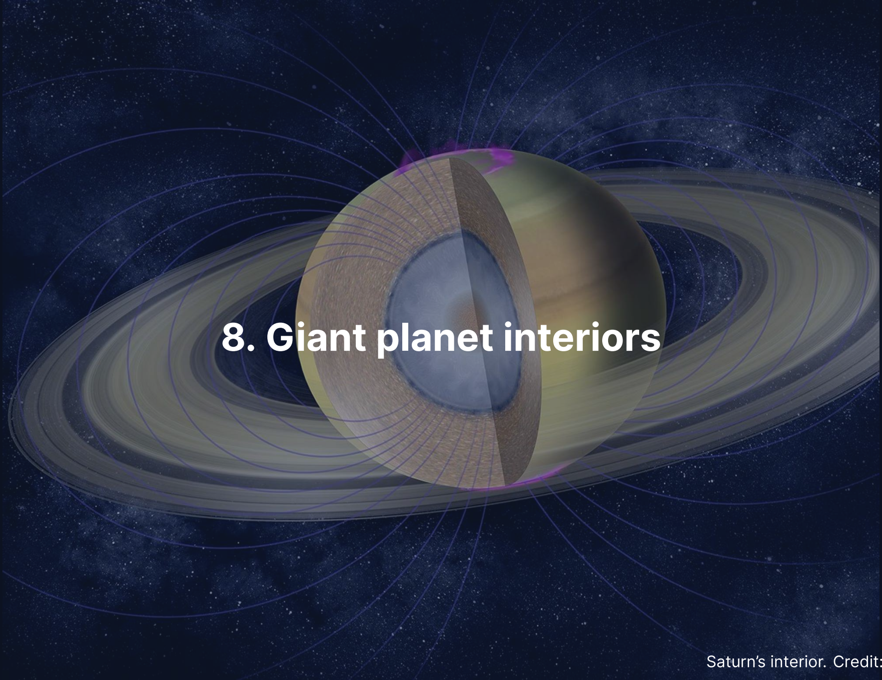
- Each shows the same architecture: crust, silicate mantle, metallic core
- Earth's two-component core (solid inner, liquid outer) is shown explicitly
- Core radius fractions span ~83% (Mercury) down to ~10–20% (Moon)

The same building blocks in very different proportions: differentiation outcomes vary widely.



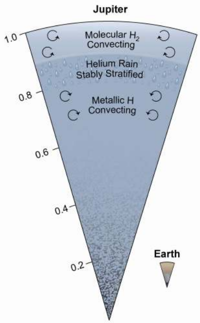
The terrestrial bodies in cross-section, at relative scale. Credit: NASA Solar System Exploration, public domain.

8. Giant planet interiors



Saturn’s interior. Credit: NASA/JPL-Caltech

Jupiter: metallic hydrogen and a dilute core

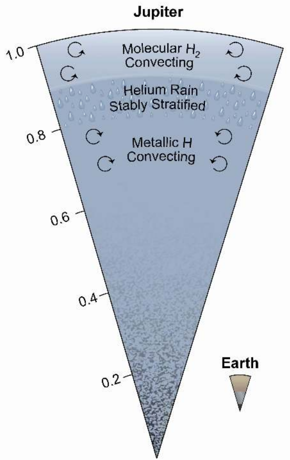


Credit: NASA/JPL-Caltech

$M = 318\,M_\oplus$, $R = 11.2\,R_\oplus$

- **Molecular** H_2 **envelope** (outer ~15%): gas → liquid
- **Metallic hydrogen** (to ~80% of R): P > 1–4 Mbar, electrons delocalised, conducts electricity
- **Dilute core:** heavy elements spread over inner ~30–50% of radius
- Central: ~4000 GPa, ~20,000 K

Radiates ~1.7× absorbed solar flux (Kelvin–Helmholtz contraction).



Jupiter’s interior: molecular and metallic hydrogen above a dilute core. Credit: NASA/JPL-Caltech/SwRI/John E. Connerney (PIA25062).

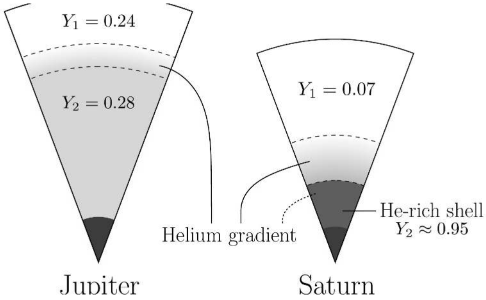
Saturn: helium rain

$M = 95\,M_\oplus$, $R = 9.4\,R_\oplus$

- Same overall structure as Jupiter, but lower pressures throughout
- Metallic hydrogen transition occurs deeper
- Larger proportional heavy-element enrichment
- **Helium rain:** as Saturn cools, He becomes immiscible in metallic H
  - Helium droplets settle toward the centre
  - Releases gravitational potential energy
  - Explains Saturn's **excess luminosity**
- Radiates ~1.8× absorbed solar flux (more than contraction alone predicts)
- **Ring seismology** (Cassini): diffuse heavy-element core to ~60% of the radius, holding ~17 $M_\oplus$ of heavy elements (Mankovich & Fuller 2021)

Helium rain is a phase separation driven by cooling, analogous to oil and water unmixing.

Helium rain, mapped

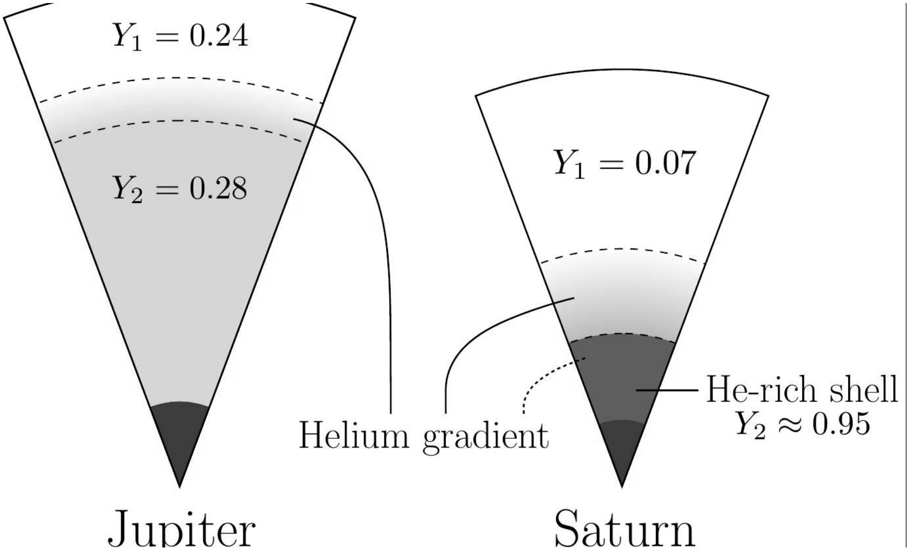


Credit: Mankovich & Fortney (2020), Fig. 3

Helium distribution in both gas giants:

- Between the dashed boundaries: the helium-gradient region created by phase separation at $P \approx 2$ Mbar
- Jupiter: a thin gradient starting near 80% of the radius; Saturn: extensive, reaching a helium-rich shell at ~35% of the radius
- The settling helium droplets release gravitational energy

Saturn's excess luminosity is gravitational energy released by sinking helium.



Helium phase separation in Jupiter and Saturn. Credit: Mankovich & Fortney (2020), Fig. 3.

9. Ice giant interiors

Uranus from Voyager 2. Credit: NASA/JPL-Caltech

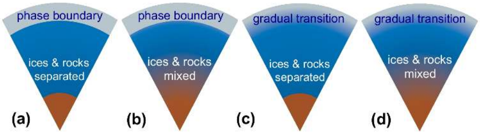
Uranus & Neptune: the canonical three layers

Canonical three-layer model:

1. **H/He envelope** (outer ~20% by mass): molecular gas → dense fluid
2. **Ice/fluid layer** (~60–70% by mass): H_2O, NH_3, CH_4 at 10–500 GPa, 2000–7000 K
   May in part reach a **superionic state** (O lattice fixed, H diffuses freely, conducts like a metal); its existence and depth range remain uncertain
3. **Rocky core** (~10–20% by mass): silicates + iron

Despite the name "ice," the material is a hot, dense, electrically conducting fluid.

Four ways to build an ice giant

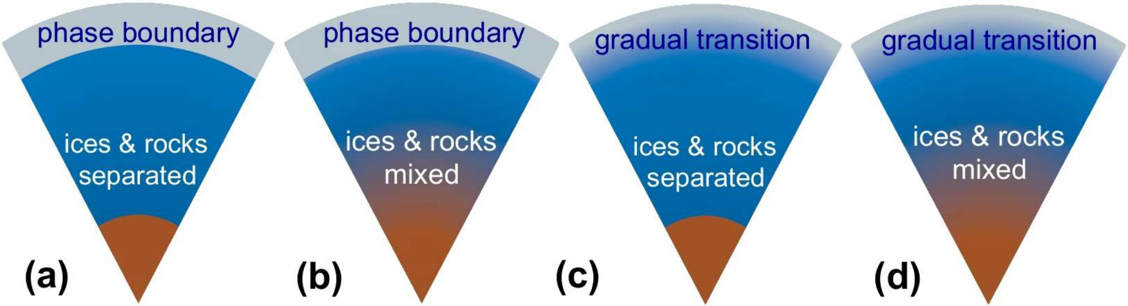


Credit: Helled et al. (2020), Fig. 4

Candidate internal structures:

- (a)–(d): layer boundaries sharp or gradual; ices and rocks separated or mixed
- Gravity data alone cannot distinguish these cases: global composition gradients remain possible
- Gradual structures would inhibit convection: a proposed explanation for Uranus's low heat flow

Whether Uranus and Neptune are differentiated is still unknown: a key driver for future missions.



Four candidate internal structures of an ice giant. Credit: Helled et al. (2020), Fig. 4.

Ice giant magnetic fields: unusual geometry

Jupiter/Saturn:

- Roughly dipolar
- Aligned with rotation axis
- Generated in deep metallic H shell

Uranus/Neptune:

- Strongly tilted (59°/47°)
- Offset from centre
- Strongly non-dipolar

The ice giant dynamos likely operate in the thin, shallow conducting ice layer rather than a deep shell, producing complex field geometries.

Uranus puzzle: essentially no measurable internal heat excess, while Neptune emits ~2.6× absorbed solar flux. Possible cause: compositional gradients inhibiting convection in Uranus.

10. Icy moon interiors & ocean worlds

Europa’s subsurface ocean. Credit: NASA/JPL-Caltech (PIA26438)

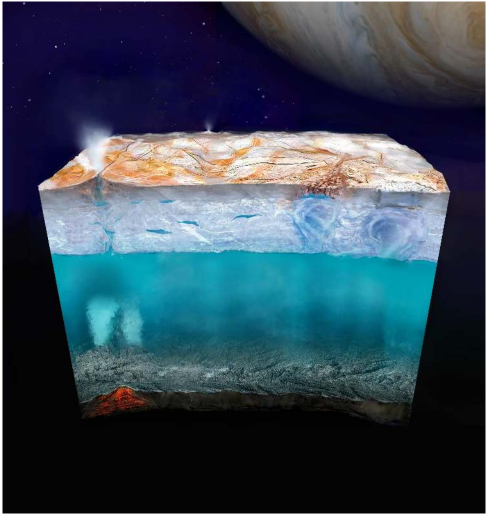
Ocean worlds: liquid water beneath ice

Several outer solar system moons harbour **subsurface oceans**:

- Maintained by **tidal heating** (frictional heat from orbital flexing) and radiogenic heat
- Insulating ice shell acts as a thermal blanket
- Prime targets for astrobiology: liquid water + chemical energy + organics

Moon	R **(km)**	ρ **(kg m^{-3})**	**Ocean evidence**
Europa	1561	3013	Magnetic induction, surface geology
Enceladus	252	1609	Geysers, libration
Titan	2575	1882	Tidal Love number k_2
Ganymede	2634	1936	Auroral oscillations (Hubble)

Europa: a habitable ocean?

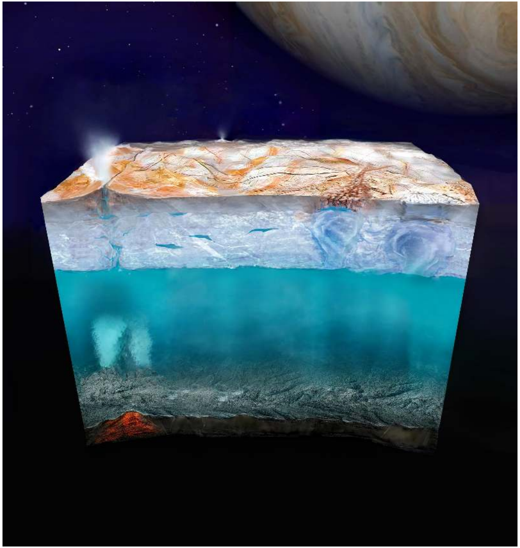


Credit: NASA/JPL-Caltech

- Ice shell: a few to a few tens of km thick
- Global ocean: roughly 100 km deep
- Rocky silicate mantle beneath the ocean

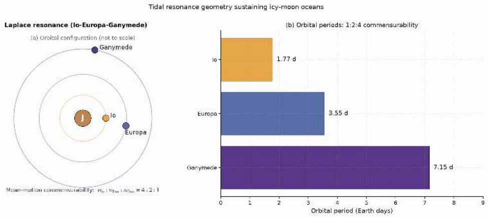
Evidence:

- Galileo: induced magnetic field (conducting layer)
- Chaos terrain: disrupted ice, like sea ice break-up
- Tidal heating from Laplace resonance (Io–Europa–Ganymede)



Europa’s subsurface environment: ice shell, global ocean, rocky seafloor. Credit: NASA/JPL-Caltech (PIA26438).

The engine of Europa’s ocean

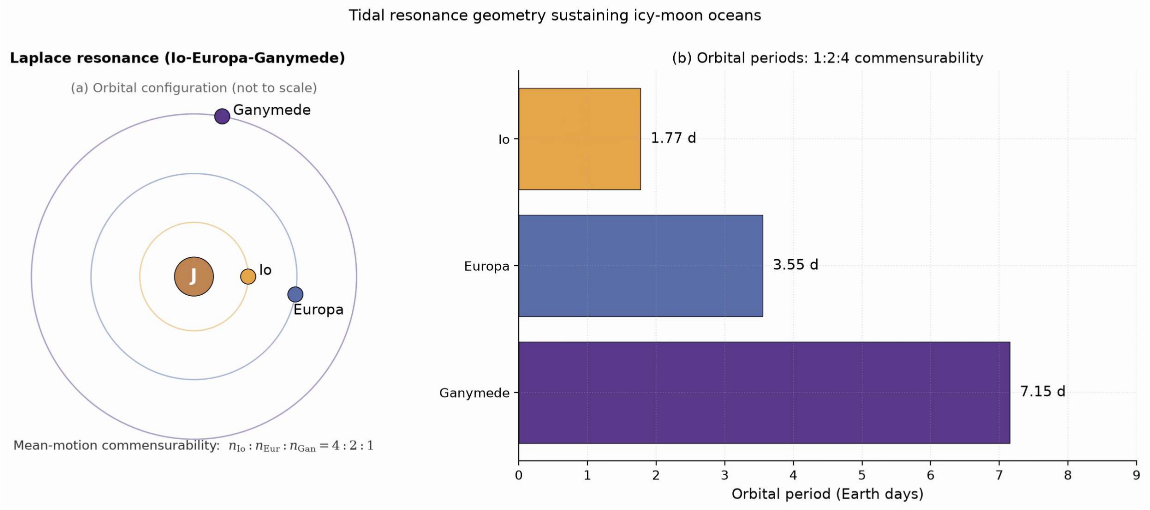


Credit: Course original; periods from NASA/JPL Solar System Dynamics

The Laplace resonance:

- Io, Europa, and Ganymede orbit with mean motions in the ratio 4:2:1 (periods 1.77, 3.55, 7.15 days)
- Repeated conjunctions force small but non-zero eccentricities against tidal damping
- The resulting time-varying tidal flexing heats Europa’s interior and keeps the ocean liquid

Without the resonance, Europa’s eccentricity would damp away and the tidal heat source would shut off.

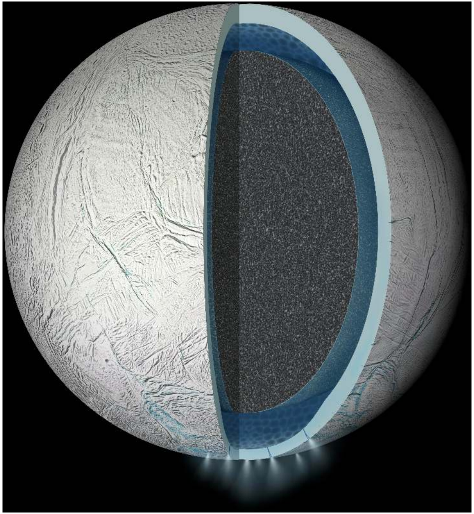


The Io–Europa–Ganymede Laplace resonance. Course original; orbital periods from NASA/JPL Solar System Dynamics.

Enceladus: direct ocean sampling

- R = 252 km, remarkably small for an active world
- Cassini flew through **geysers** erupting from “tiger stripe” fractures near the south pole
- Plume composition:
  - Water vapour and ice particles
  - Salts ($NaCl$, $NaHCO_3$)
  - Silica nanoparticles ⇒ hydrothermal activity (>90°C on seafloor)
  - Molecular H_2: potential energy source for chemosynthetic life
- Global subsurface ocean confirmed by libration measurements
- Maintained by tidal heating from resonance with Dione

A small moon with a global ocean

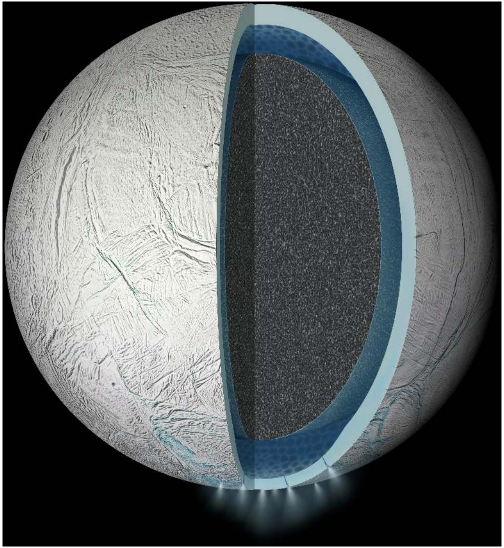


Credit: NASA/JPL-Caltech (PIA19656)

Enceladus in cutaway:

- A global liquid water ocean sandwiched between the icy crust and a porous rocky core
- Plume sources at the south polar terrain; venting first reported by Porco et al. (2006)
- The south-polar terrain radiates ~16 GW of endogenic heat (Howett et al. 2011)

Despite its 252 km radius, Enceladus sustains a global ocean and active venting.



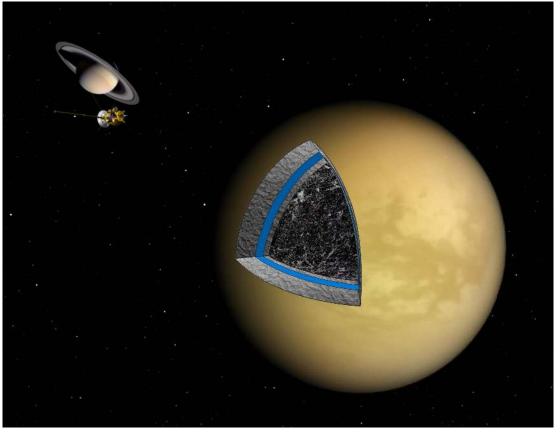
Enceladus in cutaway: ice shell, global ocean, porous rocky core. Credit: NASA/JPL-Caltech (PIA19656).

Titan: ocean beneath a thick shell

- R = 2575 km, ρ = 1882 kg m^{-3}
- ~50:50 rock–ice composition
- **Subsurface water–ammonia ocean** beneath ~50–100 km ice shell
- Evidence: Cassini Doppler tracking measured a large tidal Love number k_2
  (a global ocean decouples the ice shell from the rocky interior)
- Dragonfly rotorcraft mission: arrival in the 2030s
- Will explore surface chemistry and prebiotic conditions

Ocean worlds demonstrate that the classical habitable zone (orbital distance for surface liquid water) is far too narrow. Tidal and radiogenic heating can maintain liquid water well beyond the snow line (the ice-condensation boundary).

Titan in cross-section

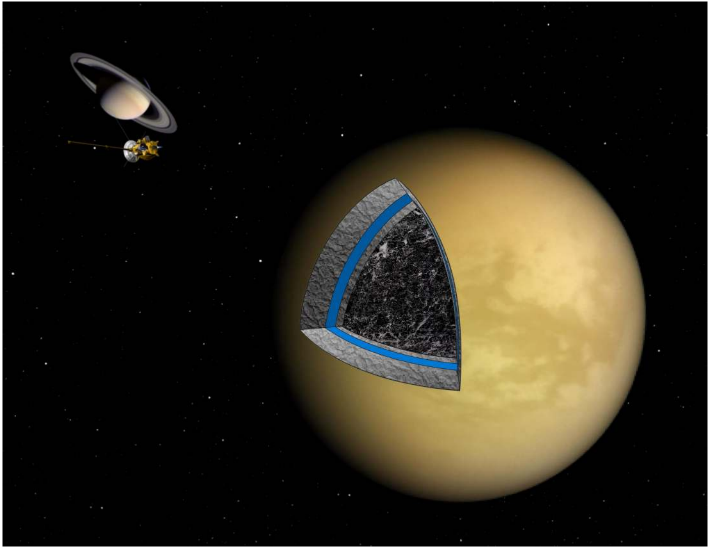


Credit: NASA/JPL

Titan's layering:

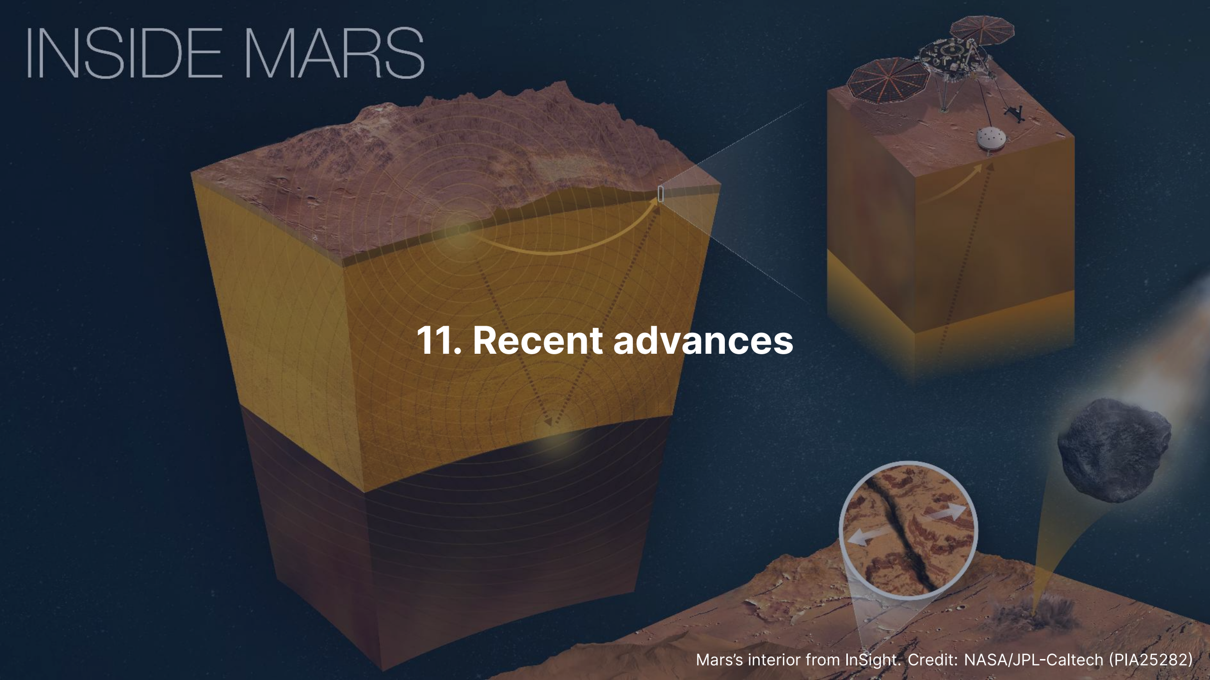
- An outer ice shell above a water–ammonia ocean
- A high-pressure ice layer below the ocean; a partially differentiated rock–ice mix in the deep interior
- The large tidal Love number k_2 from Cassini Doppler tracking shows the shell is decoupled from the interior

Titan's ocean is sealed between two ice layers, above and below.



Titan’s interior from Cassini gravity and radio science. Credit: NASA/JPL.

11. Recent advances



Mars's interior from InSight. Credit: NASA/JPL-Caltech (PIA25282)

The mass–radius diagram

A single diagnostic plot for solar system planets and exoplanets.

- Rocky planets: $R \propto M^{1/3.7}$ (bulk composition roughly similar)
- Ice/water worlds: shifted to larger radius at given mass
- Gas-envelope worlds: even larger R; Neptunes and sub-Neptunes
- Gas giants: maximum R near 1 Jupiter radius, nearly independent of mass
- Exoplanet surveys: thousands of measurements populate the diagram
- **Radius valley** at 1.5–2 $R_\oplus$: gap between super-Earths and mini-Neptunes

Source: Zeng et al. (2019); Fulton et al. (2017)

Inferring composition from mass and radius

Density alone is highly degenerate; mass and radius together constrain composition.

- Earth ($1\ M_\oplus$, $1\ R_\oplus$): average ρ = 5515 kg/m^3, rock + Fe core
- A pure "iron planet" at $1\ M_\oplus$ would have $R \approx 0.75\ R_\oplus$
- A pure "water world" at $1\ M_\oplus$ would have $R \approx 1.3\ R_\oplus$
- Intermediate compositions trace curves between these extremes
- With measurement errors, compositions are often ambiguous (rock vs. mixed)
- Tidal Love number k_2 can break degeneracies: probes internal rigidity

Source: Valencia et al. (2007); Unterborn & Panero (2019)

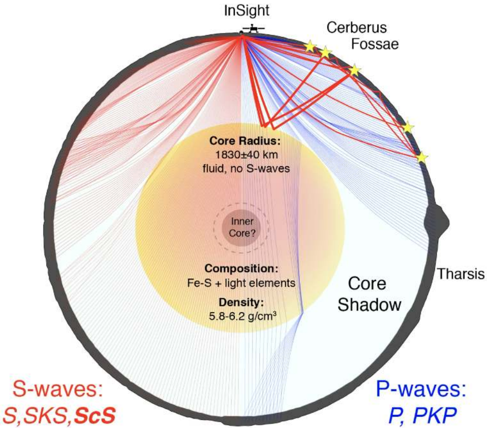
InSight: first seismology on Mars since Apollo

NASA InSight (2018–2022) delivered landmark results:

- Detected hundreds of marsquakes with the SEIS instrument
- **Liquid iron-alloy core:** R_c ~ 1830 km (Stähler et al. 2021), revised to ~1650–1675 km beneath a molten silicate layer (Khan et al. 2023; Samuel et al. 2023)
- **Thick lithosphere** (rigid outer shell): ~400–600 km (Khan et al. 2021)
- **Crust:** 24–72 km thick (thin in north, thick in south)
- Confirmed that Mars is differentiated but has a proportionally larger, less dense core than expected from simple two-layer models

First seismological constraints on another planet's interior since Apollo (1969–1977).

One planet, one great-circle cut

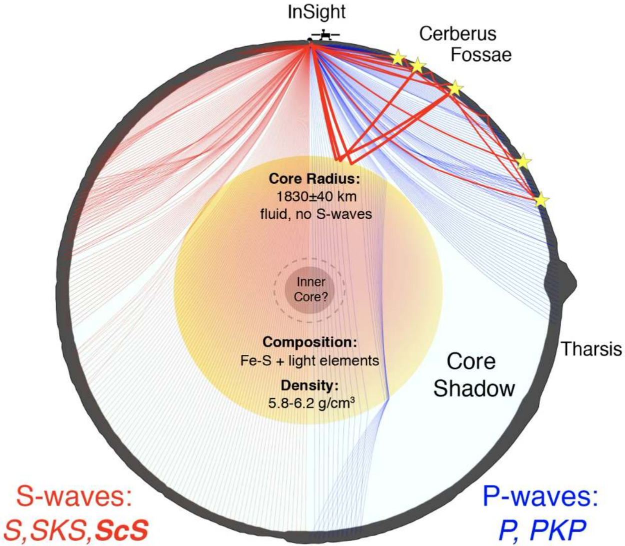


Credit: Stähler et al. (2021), Fig. 3

Mars in cross-section:

- The great circle from InSight through Olympus Mons, with the core-induced P- and S-wave shadow zones
- The seismic discontinuity at 1830 ± 40 km was interpreted in 2021 as the core–mantle boundary
- 2023 reanalyses with far-side quakes place a ~150 km molten silicate layer above a smaller, denser iron core (~1650–1675 km)

The 2021 “core surface” is now interpreted as the top of a molten rock layer; the metal core beneath is smaller and denser.



Mars in cross-section from InSight to Olympus Mons. Credit: Stähler et al. (2021), Fig. 3.

Reading a marsquake waveform

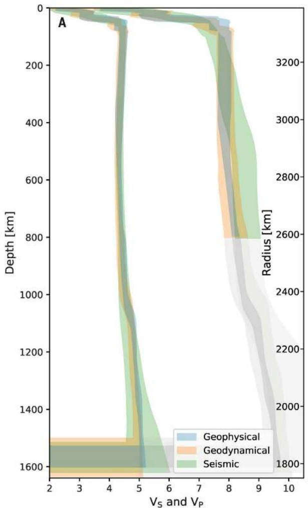
InSight's SEIS instrument revealed Mars's interior through wave-by-wave analysis.

- **P-wave arrival:** first; travels ~8 km/s through the mantle
- **S-wave arrival:** second, ~4 km/s; absent below the CMB (liquid core)
- **S-wave to P-wave delay:** measures source distance (~8 s → ~1000 km epicentre)
- Core reflections (ScS): bounce off the CMB; constrain core radius
- Mantle surface waves: constrain crustal thickness and attenuation
- Largest marsquake: **S1222a, May 2022, M 4.7**; clearly identified core phases

Every quake is a probe. InSight's ~1300 events together built a radial model of Mars's interior.

Source: Giardini et al. (2020); Duran et al. (2022) on S1222a

Mars's velocity profiles

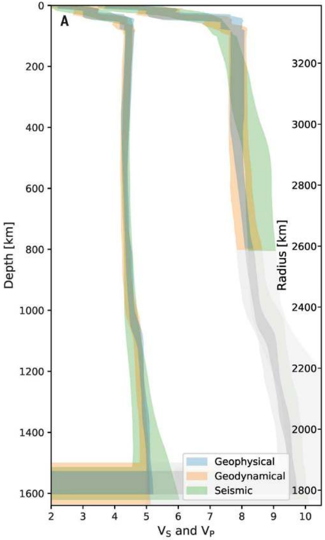


Credit: Stähler et al. (2021), Fig. 2A

v_P and v_S with depth:

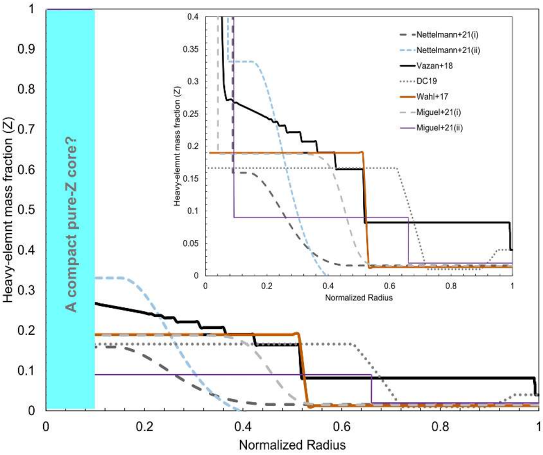
- Three independent inversions (geophysical, geodynamical, seismic) with 95% credible bands
- v_S vanishes at the core boundary: a liquid Fe-S core; v_P steps down because the liquid has a smaller bulk modulus
- Below ~800 km depth the v_P curves are extrapolations, not data-driven inversions

The bands widen where the data run out: the uncertainty is part of the result.



Mars v_P and v_S profiles from InSight. Credit: Stähler et al. (2021), Fig. 2A.

Juno: Jupiter's dilute core



Credit: Helled et al. (2022), Fig. 2, arXiv:2202.10041

- Even and odd gravity harmonics measured to high accuracy
- Heavy elements spread through the inner 0.3–0.5 R_J
- **Dilute core**: $Z(r)$ falls off gradually
- A pure core of 1–5 $M_\oplus$ is allowed, but not required
- Challenges standard core accretion (Lecture 2)
- Explanations: primordial gradient; erosion; giant impact

Ganymede: the largest moon

Ganymede (R = 2634 km, larger than Mercury) has a complex differentiated interior.

- Four layers: solid iron inner core, liquid iron outer core, silicate mantle, ice shell/ocean
- Subsurface ocean: ~100 km deep, beneath an ice shell ~150 km thick
- Ocean sandwiched between ice layers of different phases (ice I above, ice III/V/VI below)
- Confirmed by Hubble UV observations of aurora (Saur et al. 2015)
- Has its own intrinsic magnetic field (from liquid iron core dynamo)
- JUICE will dedicate most of its mission to mapping Ganymede

Source: Anderson et al. (1996); Saur et al. (2015)

Superionic ice: the ice giants' exotic interior

At the temperatures and pressures inside Uranus and Neptune, water ice adopts an extraordinary phase.

- **Superionic ice:** oxygen atoms form a rigid crystalline lattice; hydrogen ions diffuse like a liquid
- Occurs at $P > 100$ GPa, $T > 2000$ K
- First confirmed experimentally by Millot et al. (2019) using laser-driven shocks
- Predicted to constitute most of the "ice" layer in Uranus and Neptune
- Electrical conductivity of superionic ice is high ⇒ supports dynamo generation
- Explains the tilted, multipolar magnetic fields of the ice giants

Source: Millot et al. (2019), *Nature*; Ravasio et al. (2021)

PREM: seismic velocities

The Preliminary Reference Earth Model gives V_p and V_s vs. depth.

- V_p (P-wave): 6–14 km/s from crust to inner core
- V_s (S-wave): 3–7 km/s in solid layers; **zero** in the outer core
- Discontinuities at the Moho (crust–mantle), 410 km, 660 km, and CMB
- Outer core S-wave gap is the key evidence for a liquid outer core
- Inner core S-wave velocity is low (~3.5 km/s) but nonzero ⇒ solid
- Combined with density, constrains composition and phase via Birch's law

Source: Dziewonski & Anderson (1981), *Phys. Earth Planet. Inter.*

Europa Clipper: characterising an ocean world

- Launched October 2024, arrival at Jupiter system 2030
- Dozens of close Europa flybys (not an orbiter)
- Key instruments:
  - Ice-penetrating radar (REASON): ice shell thickness
  - Magnetometer: ocean salinity and depth
  - Gravity science: internal mass distribution
  - Mass spectrometer: plume/surface composition

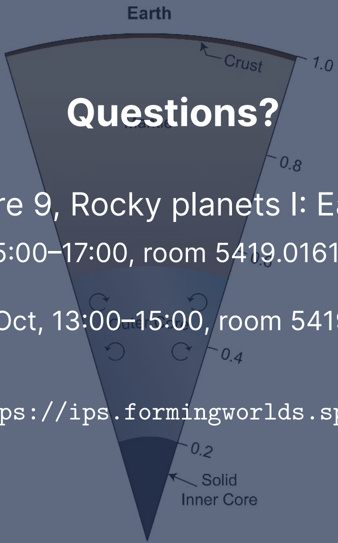
Europa Clipper will provide the most detailed constraints yet on the internal structure and habitability of an ocean world.

Summary

1. Seismology, gravity, and C/MR^2 are the primary tools for probing interiors
2. $C/MR^2 = 2/5$ for uniform sphere; lower values indicate differentiation
3. Hydrostatic equilibrium + equations of state determine interior profiles
4. Earth: crust, mantle (with phase transitions at 410, 660 km), liquid outer core, solid inner core
5. Mantle is solid but convects ($\mathrm{Ra} \sim 10^7$–10^8), driving plate tectonics
6. Giant planets: molecular → metallic H, dilute cores (Jupiter), helium rain (Saturn)
7. Ice giants: superionic water, tilted non-dipolar magnetic fields
8. Ocean worlds (Europa, Enceladus, Titan) expand the concept of habitability

Looking ahead to Lecture 9

- This lecture built the interior toolkit: seismology, C/MR^2, equations of state, and the survey from Mercury to the ocean worlds
- Lecture 9 turns to Earth and Venus: nearly equal in size and bulk composition, radically different in outcome
- Today's mantle convection and dynamo physics set up the central questions: why does Venus lack plate tectonics and a magnetic field?



Questions?

Next: Lecture 9, Rocky planets I: Earth & Venus

Mon 5 Oct, 15:00–17:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 2 Oct, 13:00–15:00, room 5419.0161 (Collegezaal)

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